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Bimetallic Tangential Rotary Cutter for Mining Machines

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(54) Title	Bimetallic tangential rotary cutter for mining machines

(57) Abstract

The invention relates to the mining industry, in particular to rotary cutters for breaking mineral and artificial materials with a wide range of strength and abrasiveness, and can be used in driving mine workings or in mineral extraction by means of shearers or roadheaders.

A bimetallic tangential rotary cutter for mining machines, comprising a body with a wearing head part reinforced with a cemented carbide insert, and a shank, wherein a hardfacing of cemented carbide material is made along the perimeter of the wearing head part of the holder, and a core made of a

material with a high value of impact toughness, characterised in that the hardfacing is made in the form of a solid toroidal shell of a high-strength material with high wear resistance, for example of tool steel, and is joined to the core by diffusion welding.

The use of the bimetallic cutter increases the hardness and, as a consequence, the wear resistance of the head part of the holder that is in contact with the rock, which leads to an increase in the service life of the cutter.

Description

The invention relates to the mining industry, in particular to rotary cutters for breaking mineral and artificial materials with a wide range of strength and abrasiveness, and can be used in driving mine workings or in mineral extraction by means of shearers or roadheaders.

A tangential rotary cutter is known (patent RU No. 44744, publ. 27.03.2005), which comprises a body, a cemented carbide insert, a supporting-rotating part and a shank. Its distinctive feature is that the ratio of the length of the head part to the length of the shank part is selected in the range 0.8... 1.2, and the radius of the cross-section of the head part increases from the cemented carbide insert towards the shank with quasi-linear intensity.

The disadvantage of this design is the rapid abrasive wear of the head part of the cutter body, which leads to exposure of the cemented carbide insert and its breaking out of the body.

A rock-breaking cutter is known (patent RU No. 2172406, publ. 20.08.2001), which comprises a working head, a cylindrical shank and a support ring fitted onto the shank. The support ring is in direct contact with the lower part of the working head of the cutter. On the mutually contacting surfaces of the cutter head and the support ring there are recesses with a depth ratio of 3–3.5, filled with an antifriction material that acts as an absorber of fines.

The disadvantage of this design is the rapid abrasive wear of the head part of the cutter body.

A composite cutter for mining machines is known (patent RU No. 119804, publ. 27.08.2012), which includes a stepped holder with three cylindrical sections, a replaceable working head placed on the front part of the holder, reinforced with a cemented carbide insert and having a cylindrical recess in its rear end surface, and a locking element for securing the working head against longitudinal displacement relative to the holder, characterised in that the locking element is made in the form of a locking set screw, which is placed in a hole threaded in its upper part, made in the side surface of the replaceable working head, and enters an annular groove in the front part of the holder.

The disadvantage is the rapid abrasive wear of the replaceable working head, which requires its frequent replacement.

A breaking or excavating tool is known (patent RU No. 2480586, publ. 27.04.2013), which comprises a body; a supporting surface including a cavity and axially protruding side walls; an insert fixed in the cavity, comprising a tip, a cone-shaped front surface, a side surface and a transition edge at the intersection of the front and side surfaces; and a ring located radially outside the protruding side walls, made of a material harder than the body.

The disadvantage of this cutter is the rapid wear of the cutter body holding the ring and the reinforcing insert of VK alloy, and their subsequent breaking out.

A tangential rotary cutter "AWR" for mining machines is known (patent RU No. 146439, publ. 10.10.2014), adopted as the prototype, which includes a holder having a body with a wearing head part reinforced with a cemented carbide insert, and a shank, characterised in that a hardfacing of cemented carbide material is made along the perimeter of the wearing head part.

The disadvantage of this cutter is the complexity of the process of applying the hardfaced layer of cemented carbide material, which requires special expensive equipment.

The technical result is an increase in the surface hardness of the cutter holder in the zone of its contact with the rock being broken, which contributes to an increase in the service life of the cutter.

The technical result is achieved in that the body with the shank is made in the form of a cylindrical shell of a high-strength material with high wear resistance, joined by diffusion welding to a core made of a high-toughness material.

The bimetallic tangential rotary cutter for mining machines is illustrated by the following figures:

- Fig. 1 — bimetallic tangential rotary cutter installed in the cutter holder of the working element of a mining machine;
- Fig. 2 — design of the bimetallic tangential rotary cutter;
- Fig. 3 — view of RSh32 tangential cutters with intensive abrasive wear of the body;
- Fig. 4 — view of rotary cutters operated after the cemented carbide insert has broken out,

where:

1. cutter;
2. cutter holder (lug);
3. working element (cutting head of a tunnelling machine);
4. retaining ring;
5. cutter body;
6. cutter shank;
7. reinforcing cemented carbide insert;
8. body shell;
9. body core.

[Figure 1 — see original PDF, page 2] Bimetallic tangential rotary cutter installed in the cutter holder of the working element of a mining machine

[Figure 2 — not reproduced in the available PDF] Design of the bimetallic tangential rotary cutter

[Figure 3 — not reproduced in the available PDF] RSh32 tangential cutters with intensive abrasive wear of the body

[Figure 4 — not reproduced in the available PDF] Rotary cutters operated after the cemented carbide insert has broken out

The bimetallic tangential rotary cutter 1 (Fig. 1) is installed in the cutter holder 2, which is fixed by welding to the working element 3 of tunnelling machines; the retaining ring 4, installed on the shank 6 of the cutter, prevents the cutter 1 from falling out of the cutter holder 2 during operation. The reinforcing cemented carbide insert 7 is installed in the cutter body 5, which is made together with the cutter shank 6 in the form of a solid cylindrical body shell 8 (Fig. 2) of a high-strength material with considerable wear resistance, joined by diffusion welding to the body core 9, which is made of a material with a high value of impact toughness.

The cutter body 5 (Fig. 1) is assembled as follows: the body shell 8 is fitted with interference onto the body core 9. The assembled structure is subjected to diffusion welding, which ensures the integrity of the bimetallic structure. The reinforcing cemented carbide insert 7 is fixed in the cutter body 5 by the cold-fit method.

During operation, the working element 3 (for example, the cutting head of a roadheader) rotates, and the reinforcing cemented carbide insert 7 together with the head part of the cutter body 5 interacts with the rock, breaking it. As a result of friction against the rock arising during cutting, the reinforcing cemented carbide insert 7 and the head part of the cutter body 5 are subject to wear. If the hardness of the material of the cutter body 5 is insufficient, the body undergoes intensive wear, accompanied by exposure of the reinforcing cemented carbide insert 7 (Fig. 3) and its breaking out (Fig. 4).

Further cutting of the rock by cutters without the reinforcing cemented carbide insert 7 is performed by the cutter body 5, as a result of which the forces acting on the tool increase up to twofold, which leads to such negative phenomena as an increase in the energy intensity of the rock-breaking process, intensification of the dynamics of operation of the working element 3, and a reduction in the productivity and service life of the tunnelling machine. Therefore, cutters without a cemented carbide insert are already regarded as failed and require immediate replacement. Thus, increasing the wear resistance of the body is an effective way of extending the service life of rotary cutters.

The use of the bimetallic cutter proposed in this application increases the hardness and, as a consequence, the wear resistance of the head part of the holder that is in contact with the rock, which leads to an increase in the service life of the cutter.

(57) Utility model claim

A bimetallic tangential rotary cutter for mining machines, comprising a body with a wearing head part reinforced with a cemented carbide insert, and a shank, characterised in that the body with the shank is made in the form of a cylindrical shell of a high-strength material with high wear resistance, joined by diffusion welding to a core made of a high-toughness material.